

RIPHAH INTERNATIONAL UNIVERSITY ISLAMABAD

Faculty of Engineering & Applied Sciences
Department of Electrical & Computer Engineering

End Term Examination — Spring 2026 Semester

Course Name:	Digital Logic Design	Course Code:	EE20203
Program:	BS Software Engineering	Section:	1
Semester:	2nd Semester	Date:	3 / June / 2026
Teacher:	Syed Hassaan	Total Marks:	40
Duration:	2-3 Hours	Student Name:	
SAP ID:		Obtained Marks:	

INSTRUCTIONS

- All 7 questions are compulsory.
- Questions 1–6 carry 5 marks each. Question 7 (Project) carries 10 marks.
- Show all working and steps clearly. Final answers without supporting work will not receive full marks.
- Write your Name and SAP ID on all answer sheets.
- Mobile phones and electronic devices are strictly not allowed.

Q1. [Quine-McCluskey Method]**[5 Marks]**

Minimize the Boolean function $F(A, B, C, D) = \sum m(0, 1, 3, 7, 8, 9, 11, 15)$ using the Quine-McCluskey (tabulation) method. Show all grouping tables, list all Prime Implicants (PIs), construct the Prime Implicant Table, identify all Essential Prime Implicants (EPIs), and write the final minimized SOP expression.

Q2. [Adders & Subtractors]**[5 Marks]**

Design a 4-bit Adder/Subtractor circuit using a 4-bit ripple carry adder and XOR gates.

Q3. [Decoders & Encoders]**[5 Marks]**

Implement the following two Boolean functions simultaneously using a single 3-to-8 line decoder (74x138) and external OR gates. Clearly show which output lines of the decoder are connected to each OR gate.

$$F1(A, B, C) = \sum m(1, 3, 5, 7)$$

$$F2(A, B, C) = \sum m(0, 2, 6, 7)$$

Q4. [Multiplexers & Demultiplexers]**[5 Marks]**

Using an 8-to-1 multiplexer (74151A), implement the Boolean function $F(A, B, C) = \sum m(0, 2, 3, 5, 7)$. Connect A, B, C to select lines S2, S1, S0 respectively and determine the value (0 or 1) to be applied to each data input I0 through I7. Draw the complete connection diagram.

Q5. [Latches & Flip-Flops]**[5 Marks]**

A JK flip-flop is connected so that $J = K = 1$. Starting from $Q = 0$, trace the output Q for 6 consecutive clock pulses and draw the timing diagram.

Q6. [Counters]**[5 Marks]**

Design a synchronous Mod-6 (count 0 to 5) up-counter using JK flip-flops.

Q7. *[Semester Project]*

[10 Marks]

Describe your semester project. Your answer must cover the following:

- Project title and a brief overview of what the circuit/system does.
 - The digital logic components and concepts used (e.g., gates, flip-flops, counters, MUX, etc.).
 - A block diagram or schematic of the overall design.
 - Any challenges faced during design/implementation and how you resolved them.
 - The practical application or real-world relevance of your project.
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— END OF PAPER —

Total: 40 Marks (Q1–Q6: 5 marks each | Q7: 10 marks) | Best of Luck!